

Group 6

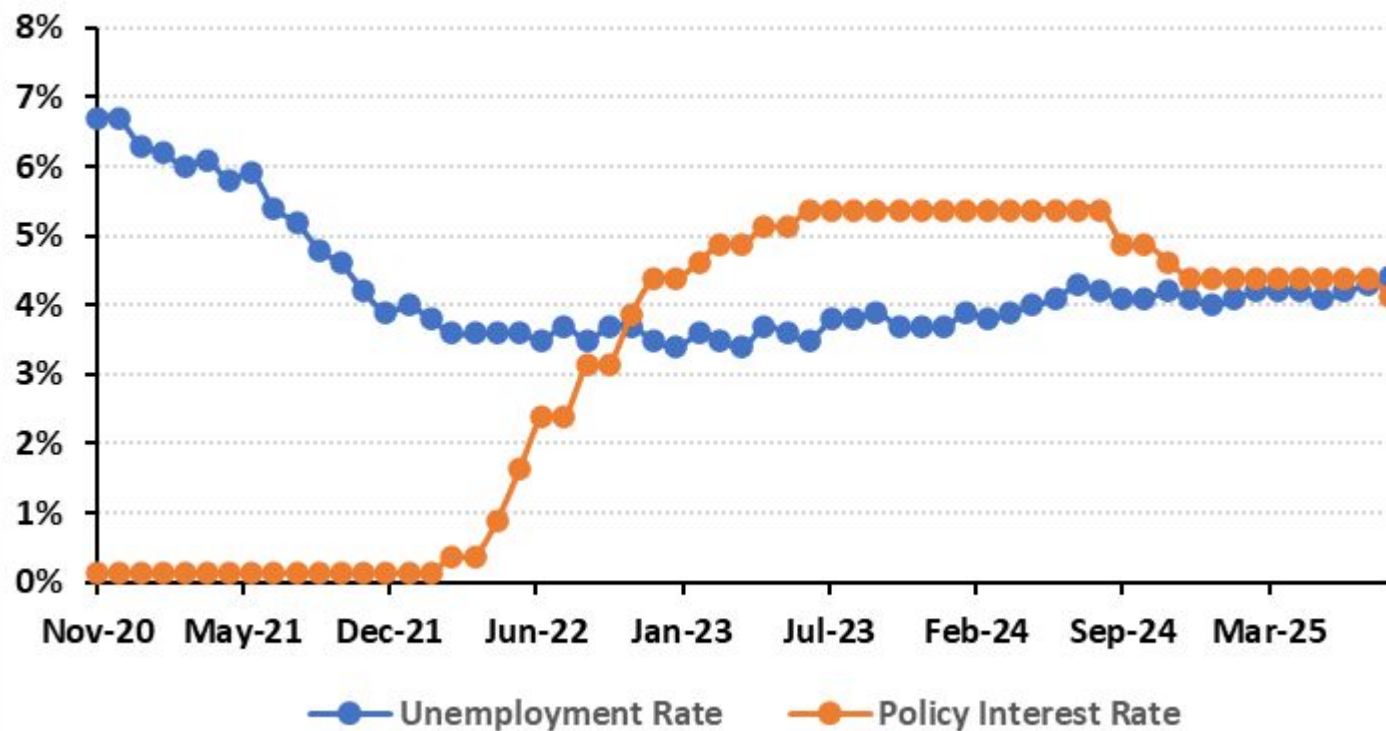
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Part 1: US Equity Portfolio

Strategy Selection

- Historical Empirical Performance:
 - From 1927 to 2017, value strategy has outperformed growth strategy.
 - Value stock has outperformed growth stock in different time frames.
- Strategy Nature:
 - Value Strategy: Negative Feedback
 - Growth Strategy: Positive Feedback
- Macroeconomics:
 - Current Downside Pressure: tariff, slow employment growth
 - Relatively high interest rate in the history

US Policy Rate & Unemployment Rate Curve



Factors and Variables

1. Earnings Yield

- $(\text{Earnings Per Share} / \text{Stock Price}) \times 100$
- A high earnings yield $\square$ substantial profits relative to its market price
- Higher yields $\square$ better returns

2. P/E Ratio

- $\text{P/E} = \text{Stock Price} / \text{Earnings Per Share}$
- Measures how much investors pay per dollar of earnings.
- Low ratios $\square$ mispriced stocks

3. EV/EBITDA

- $\text{EV/EBITDA} = (\text{Market Cap} + \text{Debt} - \text{Cash}) / \text{EBITDA}$
- Enterprise Value to EBITDA
- Assesses total company value against operational earnings
- Low ratios $\square$ ideal for comparing firms with varying debt levels in a high-interest-rate era.

4. **P/S Ratio**

- $P/S = \text{Stock Price} / \text{Sales Per Share}$
- Evaluate price relative to revenue
- Low P/S $\square$ market undervalues the company's sales generation

5. **P/B Ratio**

- $P/B = \text{Stock Price} / \text{Book Value Per Share}$
- Compares market price to net asset value.
- Low P/B $\square$ stocks trading below their book value
- Emphasizes asset-based valuation

6. **FCF Yield**

- $\text{FCF Yield} = (\text{Free Cash Flow} / \text{Market Capitalization}) \times 100$
- Cash generation after expenses as a percentage of market cap.
- High yields highlight companies with strong and distributable cash flows
- Focuses on real cash

Portfolio Construction

Portfolio Philosophy: To construct a safe and consistent active portfolio

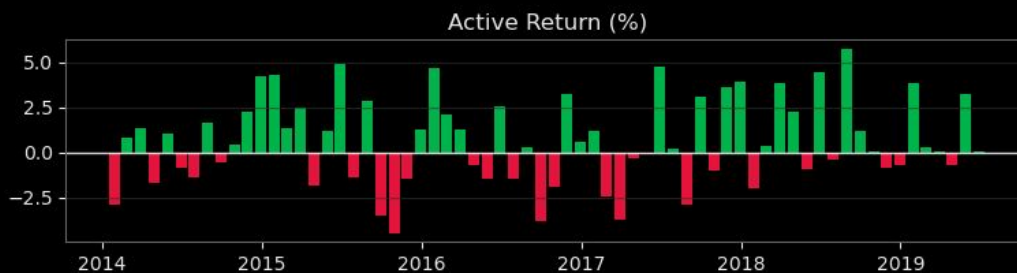
0. What factors to use? 
1. What are the best factor weights?
2. What is the best portfolio size?
3. Is this portfolio consistent?

1. What are the best factor weights?

1. 10 Years of data (5 years of train & test)
2. Scaling factors to z-score
3. Randomly searching with Dirichlet distribution
4. $\text{Score} = \sum \text{Weight} * \text{Factor}$
5. Ranking universe based on “Score”
6. Picking top 45 names
7. Comparing active return to benchmark
8. Repeated for 10,000 portfolios
9. Averaging top 50 portfolios factor weights
10. Similar steps for holding weights

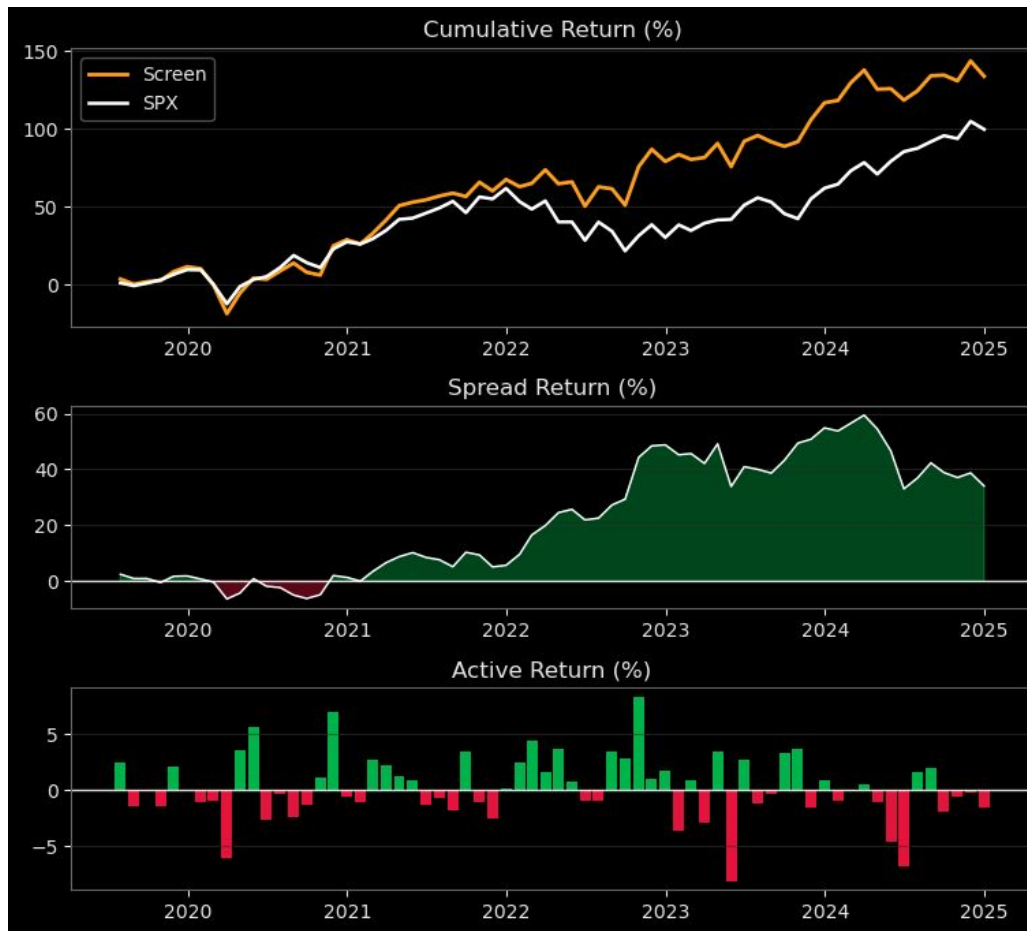
AVG Factor Weights :
P/E: 26.42
P/B: 6.29
P/S: 7.37
EV/EBITDA: 38.14
FCF Yield: 4.86
Earnings Yield: 16.91

Training Period



Total Return (%)	136.92
Benchmark Total (%)	59.15
Mean Return (%)	1.39
Mean Active (%)	0.63
Min Return (%)	-9.87
Max Return (%)	11.70
Std Dev (%)	3.93
Semi variance	17.35
Tracking Error (%)	2.44
Sharpe Ratio	1.17
Up Capture (%)	122.58
Down Capture (%)	73.78
Jensen Alpha (%)	8.60
Information Ratio	0.89
Beta	0.91
Correlation	0.79

Testing Period



Total Return (%)	134.00
Benchmark Total (%)	99.94
Mean Return (%)	1.47
Mean Active (%)	0.29
Min Return (%)	-18.60
Max Return (%)	17.68
Std Dev (%)	5.99
Semi variance	33.37
Tracking Error (%)	2.92
Sharpe Ratio	0.74
Up Capture (%)	103.79
Down Capture (%)	87.56
Jensen Alpha (%)	3.00
Information Ratio	0.35
Beta	1.05
Correlation	0.87

Portfolio Holdings (=45)

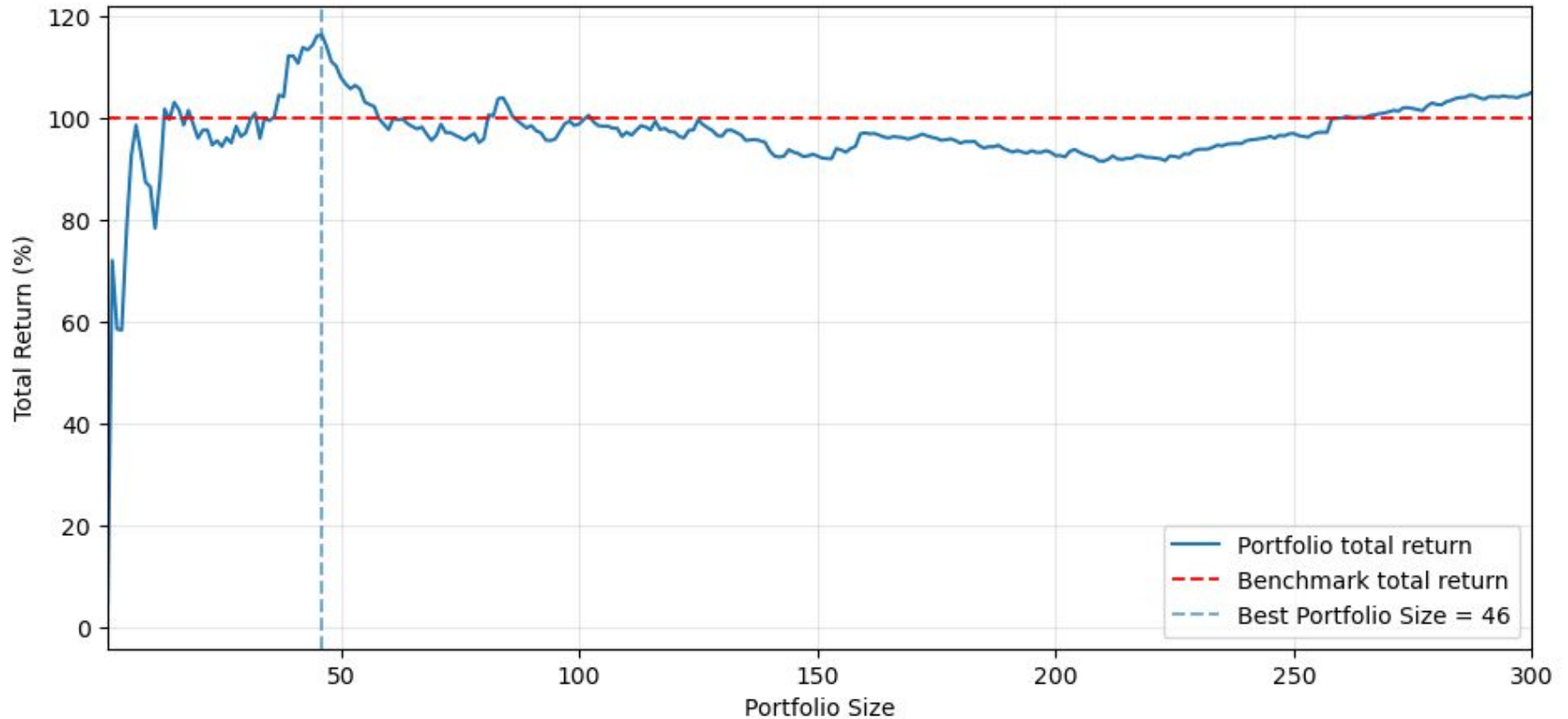
	Ticker	Weight
0	LULU	16.37
1	UHS	7.29
2	FISV	5.62
3	ACGL	5.51
4	MOH	5.24
5	ELV	4.87
6	TROW	4.68
7	FANG	4.65
8	PGR	4.26
9	OMC	3.71
10	CI	3.27
11	DHI	3.24
12	GIS	2.91

13	ALL	2.86
14	DECK	2.58
15	AIG	2.51
16	CF	1.89
17	DAL	1.41
18	LKQ	1.40
19	MOS	1.27
20	DVN	1.21
21	TRV	1.18
22	COP	1.17
23	VZ	0.98
24	CAG	0.98
25	LEN	0.94
26	CMCSA	0.87

27	GPN	0.84
28	CHTR	0.69
29	TGT	0.69
30	OKE	0.57
31	UAL	0.51
32	EOG	0.43
33	MET	0.43
34	EIX	0.42
35	T	0.41
36	AIZ	0.38
37	APA	0.35
38	CINF	0.34

39	GL	0.30
40	EMN	0.30
41	PHM	0.23
42	BG	0.10
43	HIG	0.09
44	CTRA	0.05

2. What is the best portfolio size?



3. Is this portfolio consistent?

Active return

slope : 0.01244 (percent per run)

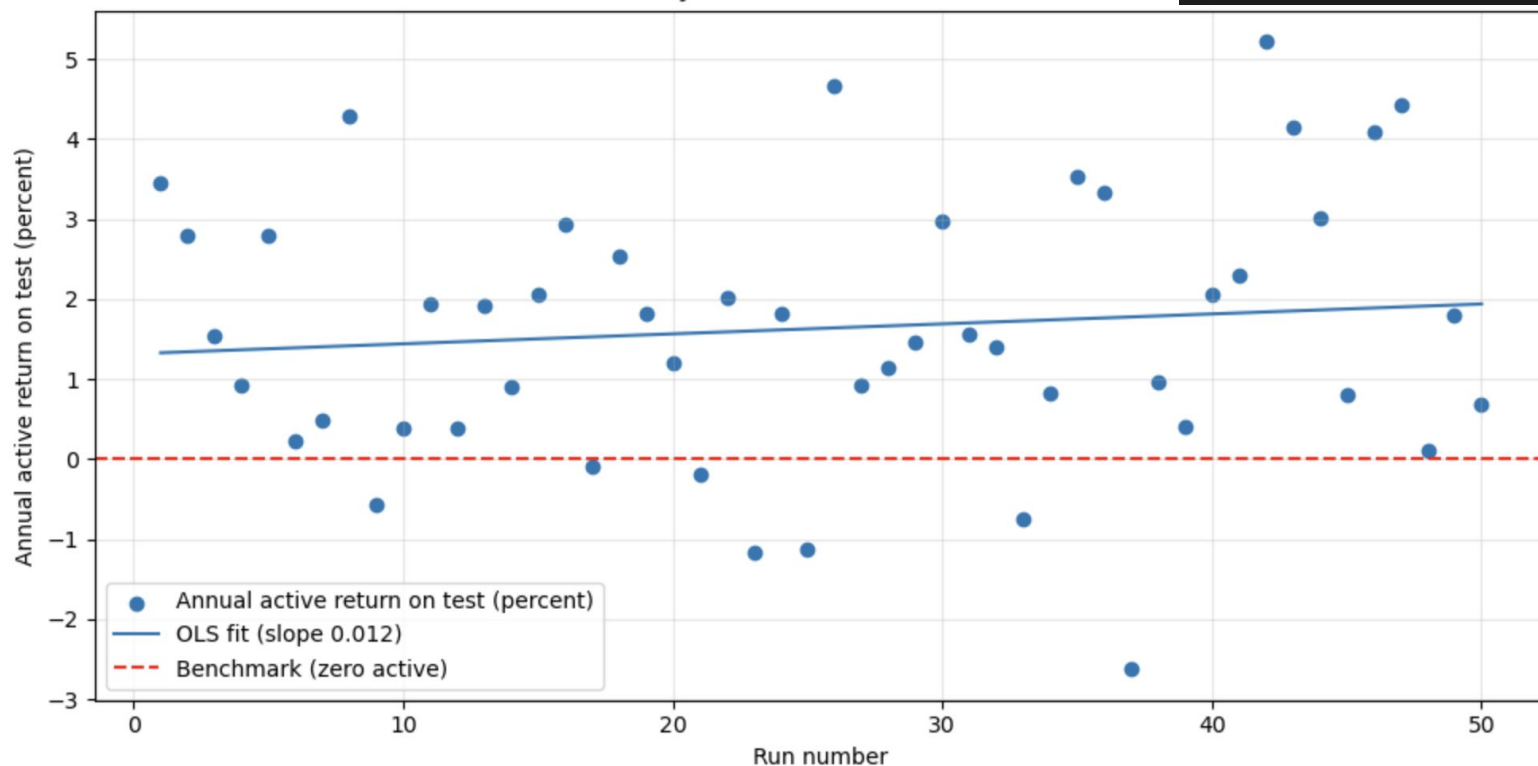
standard error : 0.01628

t statistic : 0.764

p value : 0.449

90% CI : [-0.01487, 0.03976]

Stability of model test active return



Part 2: Canadian Equity Portfolio

Selection Process

Morningstar Universe

- Due to blue heaven being an endowment fund, we operate under the assumption that the fund has a long-term horizon and is looking to get the most return while being exposed to limited risk
- For selection of the three managers for Canadian equity, we looked at the universe of Canadian Funds for the equity, dividend and income, and small cap categories on Morningstar.
- We prioritized Series I and F funds.

Morningstar Direct -- Open End Funds

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☐	Rel	(	Field Name	Operator	Value
☐ 1	---		Domicile	=	Canada
☐ 2	And		Morningstar Category	=	Canadian Small/Mid Cap Equity

☐ Name	Morningstar Rating Overall	Morningstar Medalist Rating	Total Ret 1 Day (Daily) CAD	Total Ret Annlzd 5 Yr (Daily) CAD	Sharpe Ratio 5 Yr (Mo-End) CAD	Information Ratio 5 Yr (Mo-End) CAD	Treyrnor Ratio 5 Yr (Mo-End) CAD	Max Drawdown CAD	
								11/01/2022 to 10/31/2025	
☐ MM Fund Series F	★★★★★	Bronze	0.46	19.63	1.11	0.18	28.30	-7.24	
☐ Guardian Canadian Focused Equity I	—	—	1.12	20.17	1.48	0.79	23.22	-6.47	
☐ EdgePoint Canadian Portfolio Series F	★★★★★	Bronze	1.04	20.86	1.44	0.92	22.59	-5.05	
☐ Guardian Canadian Focused Equity F	★★★★★	Neutral	1.12	—	1.44	0.69	22.47	-6.60	
☐ DFA Canadian Vector Equity Class I	—	—	0.84	21.19	1.41	1.14	20.27	-7.18	

Z Score

- We expanded our manager evaluation by adding three risk-adjusted performance metrics over the trailing 5-year period: Sharpe Ratio, Treynor Ratio, and Information Ratio.
- Standardization is necessary because the three ratios measure different forms of risk-adjusted performance and have different magnitudes.
- we averaged the Z-scores for each fund to produce a composite risk-adjusted performance score, allowing us to rank the funds objectively. We then selected the top 3–5 managers based on this composite score

Name	Information Ratio 5 Yr (Mo-End)	Sharpe Ratio 5 Yr (Mo-End)	Treynor Ratio 5 Yr (Mo-End)	Drawdown 2022-11-0	Information Ratio Score	Sharpe Ratio Score	Treynor Ratio Score	Overall Score
Brandes Canadian Equity I	0.74	1.70	32.86	-3.07	1.99	2.15	1.68	1.92
Brandes Canadian Equity Class F	0.57	1.60	30.50	-3.27	1.72	1.92	1.47	1.69
McElvaine Value F	-0.05	1.22	36.48	-8.33	0.76	1.11	1.99	1.27
Pender Small Cap Opportunities F	0.05	0.79	16.79	-13.93	0.91	0.19	0.28	0.46
Chou RRSP Class F	-0.11	0.75	19.17	-14.22	0.67	0.12	0.49	0.42
CI North Amer Sm/Mid Cap Eq Fd(SerEF U)	-0.63	0.61	10.42	-12.22	-0.13	-0.18	-0.27	-0.19
CI North Amer Sm/Mid Cap Eq Fd (Ser F U)	-0.64	0.61	10.31	-12.28	-0.15	-0.19	-0.28	-0.20
CI North Amer Sm/Mid Cap Eq Cl(SerEFShrs)	-0.64	0.61	10.27	-12.36	-0.16	-0.19	-0.29	-0.21
CI North Amer Sm/Mid Cap Eq Cl(SerEShrs)	-0.76	0.54	9.02	-12.87	-0.34	-0.33	-0.40	-0.35
Dynamic Small Business Series I	-1.07	0.49	7.98	-12.16	-0.83	-0.44	-0.49	-0.58
Canada Life Canadian Fcs Sm-Md Cp Fd F	-1.11	0.41	6.55	-8.78	-0.88	-0.61	-0.61	-0.70
Dynamic Small Business Ser FI	-1.23	0.40	6.22	-12.91	-1.06	-0.64	-0.64	-0.77
Dynamic Small Business Series F	-1.23	0.39	6.11	-12.95	-1.07	-0.65	-0.65	-0.78
Canada Life Canadian Fcs Sm-Md Cp Fd W	-1.24	0.33	5.00	-9.50	-1.08	-0.78	-0.75	-0.86
Goodwood Capital F	-0.77	0.00	-4.00		-0.35	-1.48	-1.53	-1.11
Mean	-0.54	0.70	13.58					
Standard Deviation	0.64	0.47	11.50					

Attribution Analysis

- In each category to run an attribution analysis on selected top 3-5 funds based on the Z score.
- Using the attribution analysis score we selected the top managers in each category based on highest active return.

Multiple Investment Attribution Total

Time Period: 11/1/2020 to 10/31/2025

	Calculation Benchmark	Allocation Effect	Selection Effect	Active Return	Benchmark Return
Brandes Canadian Equity Class F	Morningstar DM Canada Small 65/35 GR CAD	8,90	56,85	65,75	113,13
McElvaine Value F	Morningstar DM Canada Small 65/35 GR CAD	19,56	33,63	53,19	113,13
Brandes Canadian Equity I	Morningstar DM Canada Small 65/35 GR CAD	8,90	56,85	65,75	113,13

Attribution Analysis

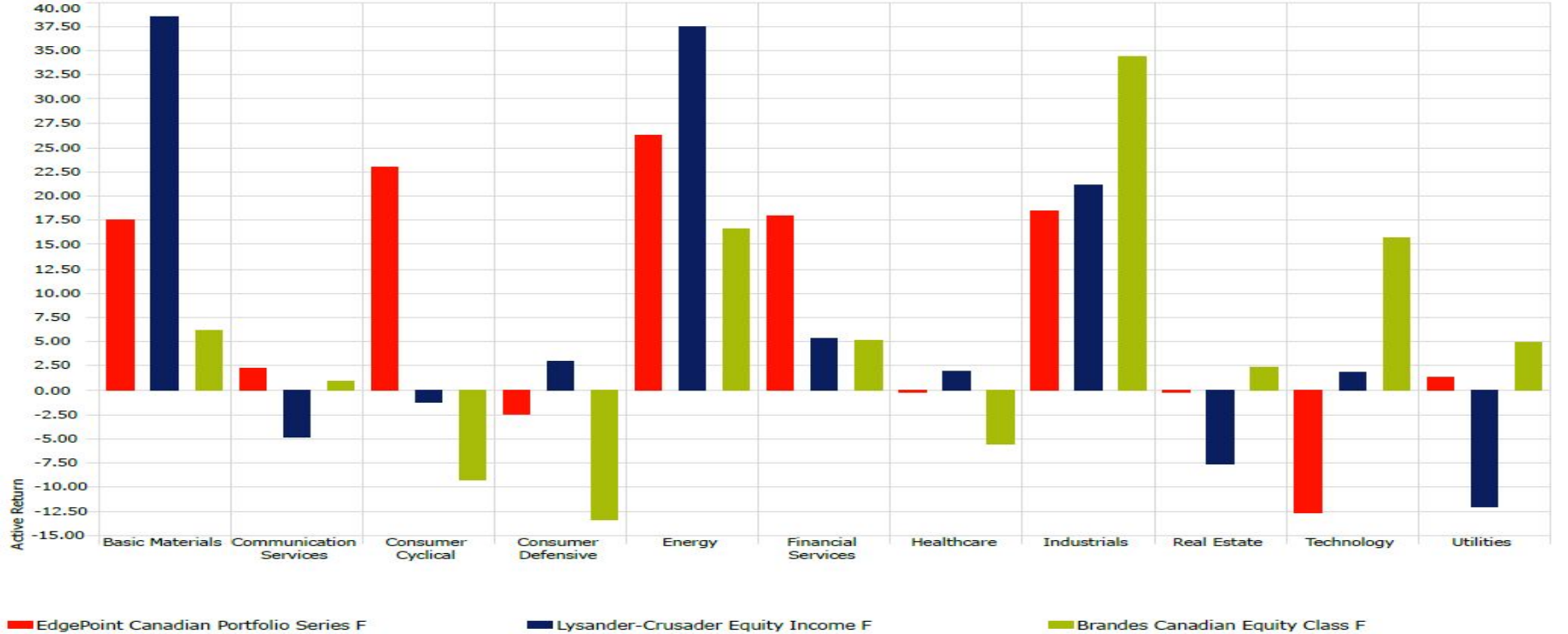
Multiple Investment Attribution Total

Time Period: 2020-11-01 to 2025-10-31

	Calculation Benchmark	Allocation Effect	Selection Effect	Active Return	Benchmark Return
EdgePoint Canadian Portfolio Series F	Morningstar Canada GR CAD	6.67	84.83	91.50	129.89
Lysander-Crusader Equity Income F	Morningstar Canada GR CAD	7.10	76.54	83.64	129.89
Brandes Canadian Equity Class F	Morningstar Canada GR CAD	-9.50	58.49	48.99	129.89

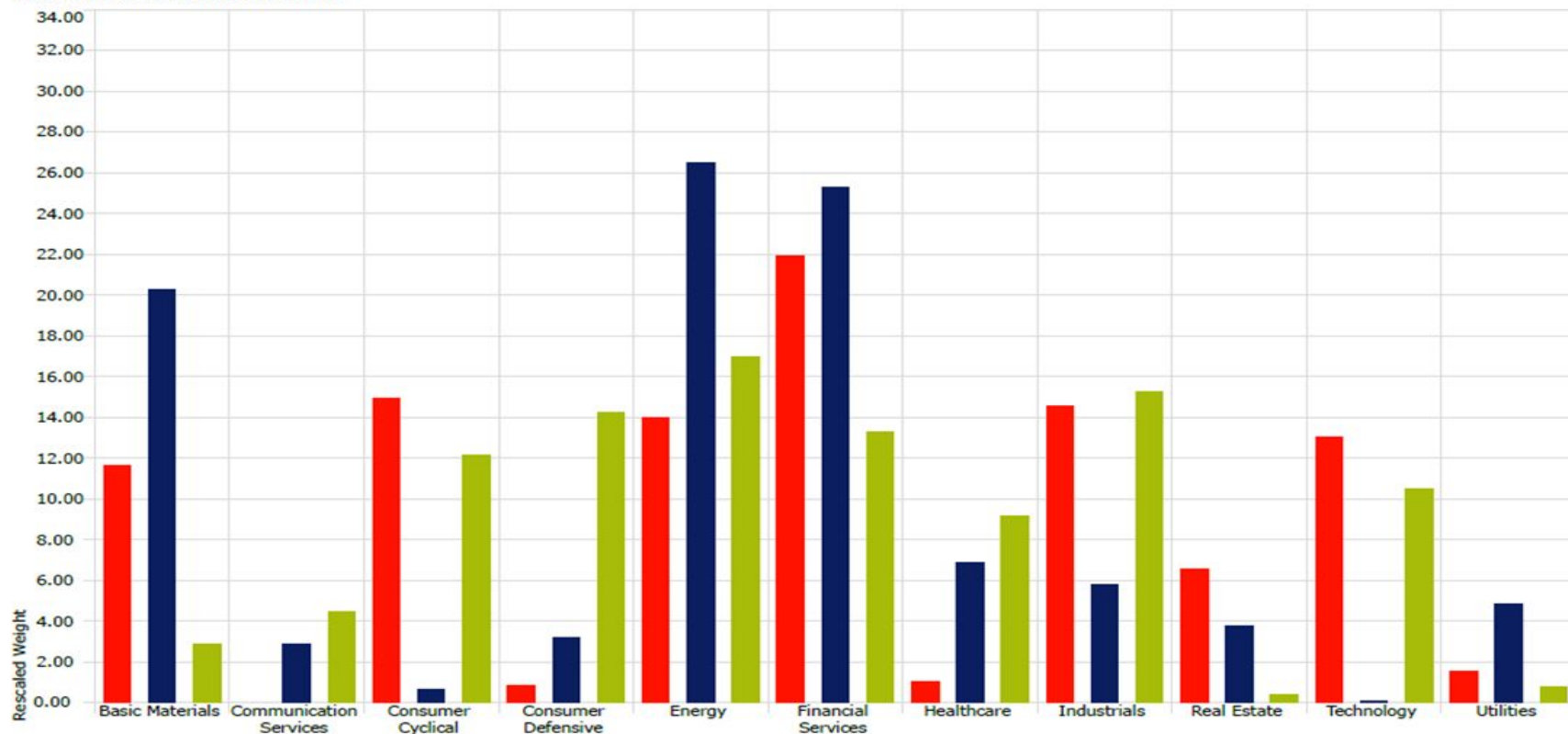
Active Return by Sector

Time Period: 2020-11-01 to 2025-10-31



Selection Effect by Sector

Time Period: 2020-11-01 to 2025-10-31



EdgePoint Canadian Portfolio Series F

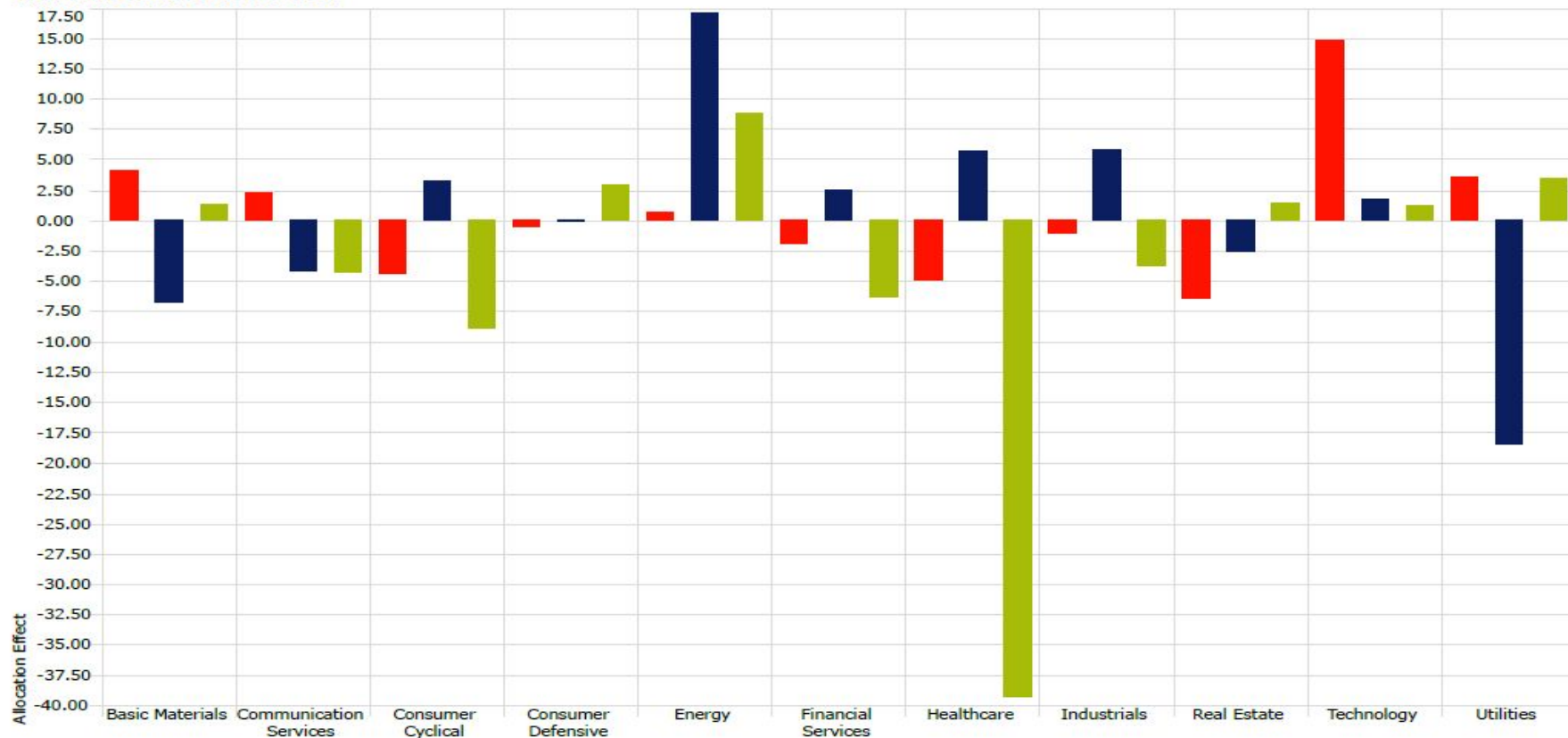
Lysander-Crusader Equity Income F

Brandes Canadian Equity Class F

Source: Morningstar Direct

Allocation Effect by Sector

Time Period: 2020-11-01 to 2025-10-31



EdgePoint Canadian Portfolio Series F

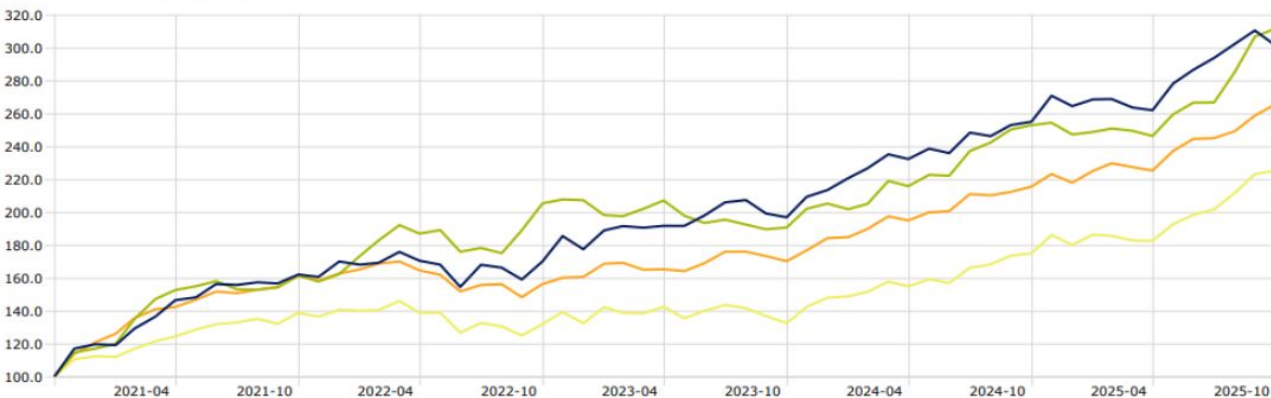
Lysander-Crusader Equity Income F

Brandes Canadian Equity Class F

Return Analysis

Investment Growth

Time Period: 2020-11-01 to 2025-10-31



EdgePoint Canadian Portfolio Series F

Lysander-Crusader Equity Income F

Brandes Canadian Equity Class F

S&P/TSX Composite TR

25.53%

Lysander-Crusader Return

24.68%

21.58%

Brandes Return

17.64%

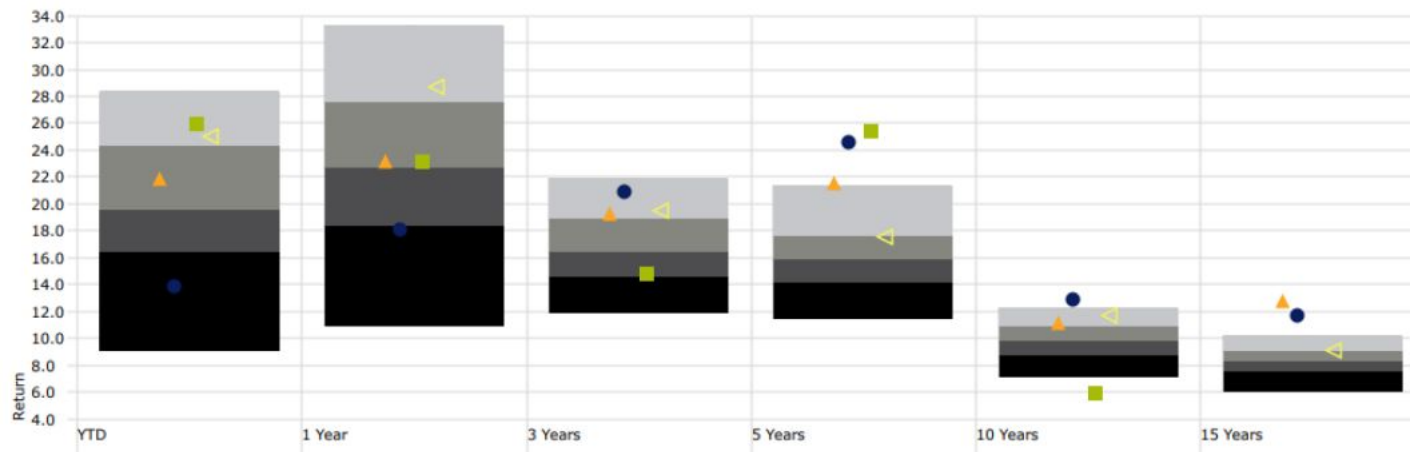
Edgepoint Return

Benchmark

Trailing Returns

Peer Group (5-95%): Funds - Canada - Canadian Equity Calculation Benchmark: Morningstar Canada GR CAD

Top Quartile 2nd Quartile 3rd Quartile Bottom Quartile



● EdgePoint Canadian Portfolio Series F

■ Lysander-Crusader Equity Income F

▲ Brandes Canadian Equity Class F

◀ S&P/TSX Composite TR

Trailing Returns

Data Point: Return Calculation Benchmark: Morningstar Canada GR CAD

	YTD	1 Year	3 Years	5 Years	10 Years	15 Years
EdgePoint Canadian Portfolio Series F	13.93	18.18	21.00	24.68	12.93	11.78
Lysander-Crusader Equity Income F	26.04	23.22	14.92	25.53	5.97	—
Brandes Canadian Equity Class F	21.90	23.23	19.33	21.58	11.25	12.82
S&P/TSX Composite TR	25.13	28.75	19.52	17.64	11.71	9.19

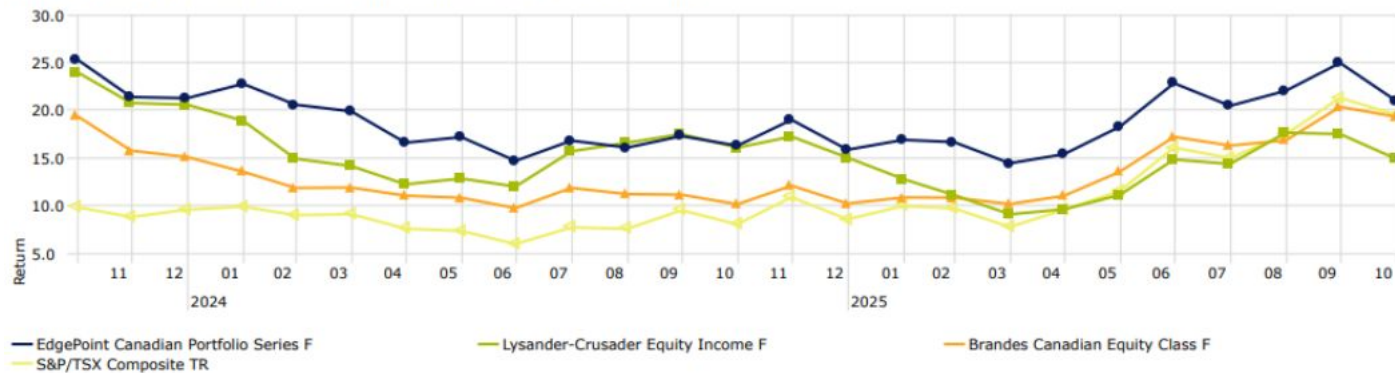
Upper
Quartile
5-year
Performance

Consistently
Top of
Category

Rolling Returns

Time Period: 2020-11-01 to 2025-10-31

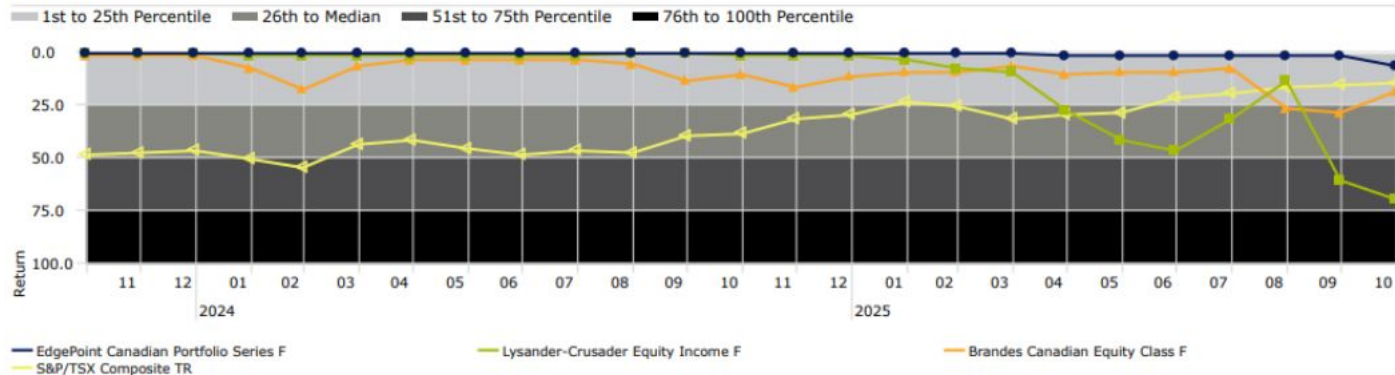
Rolling Window: 3 Years 1 Month shift Calculation Benchmark: Morningstar Canada GR CAD



Rolling Returns (Descending Rank)

Time Period: 2020-11-01 to 2025-10-31

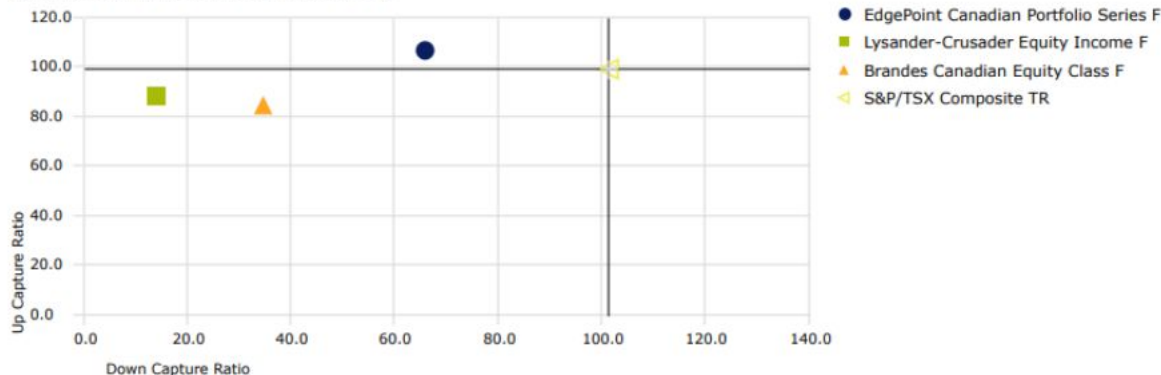
Rolling Window: 3 Years 1 Month shift Calculation Benchmark: Morningstar Canada GR CAD



Upside-Downside Capture

Time Period: 2020-11-01 to 2025-10-31

Calculation Benchmark: Morningstar Canada GR CAD



106.34%

Upcapture for Edgepoint

Market Performance

Time Period: 2020-11-01 to 2025-10-31

Calculation Benchmark: Morningstar Canada GR CAD

	Up Period Percent	Down Period Percent	Best Month	Worst Month	Best Quarter	Worst Quarter	Up Capture Ratio	Down Capture Ratio	R2
EdgePoint Canadian Portfolio Series F	63.33	36.67	17.07	-8.05	14.54	-12.12	106.34	65.94	84.37
Lysander-Crusader Equity Income F	68.33	31.67	14.76	-6.95	25.76	-8.41	88.21	13.87	55.65
Brandes Canadian Equity Class F	75.00	25.00	13.95	-6.27	16.65	-10.71	84.56	34.65	81.12
S&P/TSX Composite TR	65.00	35.00	10.57	-8.71	12.50	-13.19	98.93	101.34	99.94

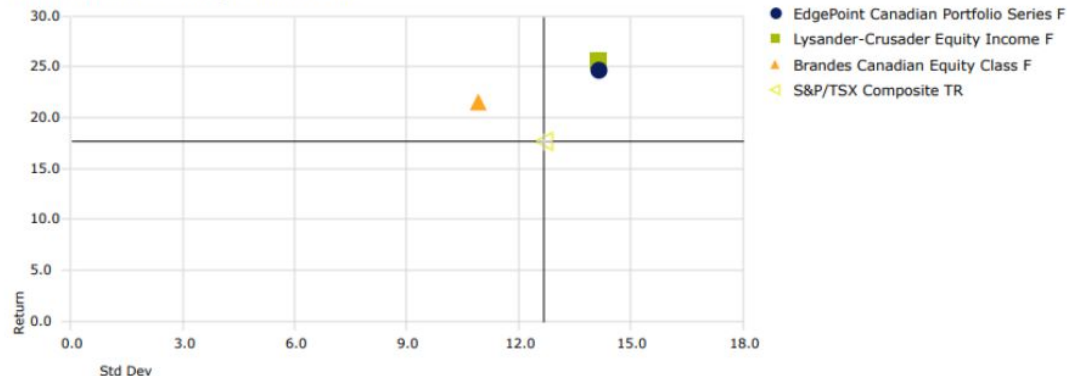
13.87%

Down Capture for
Lysander-Crusader

Risk-Reward

Time Period: 2020-11-01 to 2025-10-31

Calculation Benchmark: Morningstar Canada GR CAD



<1.00

All 3 of the firms betas

Lysander

Edgepoint

Brandes

TSX

1.50

1.44

1.58

1.05

Sharpe Ratio

Max Drawdown

Lysander -08.86%

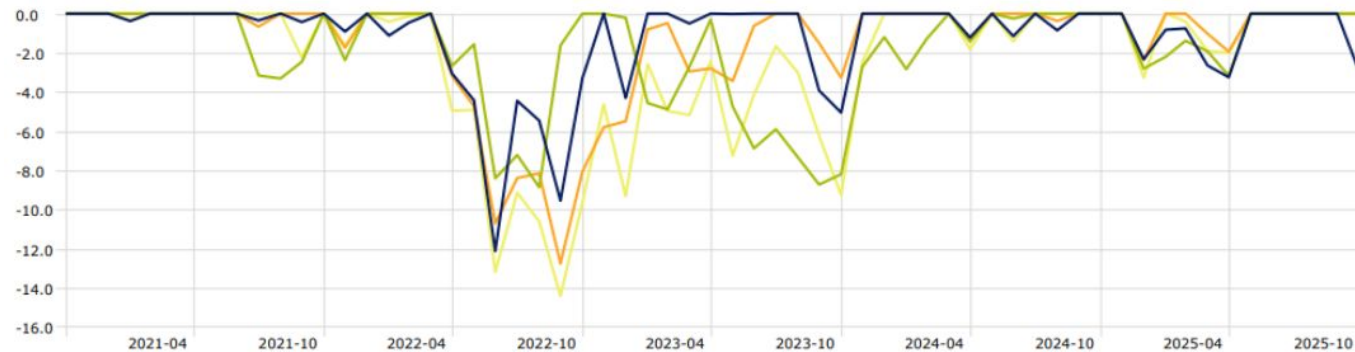
Edgepoint -12.12%

Brandes -12.75%

TSX -14.41%

Drawdown

Time Period: 2020-11-01 to 2025-10-31



EdgePoint Canadian Portfolio Series F
S&P/TSX Composite TR

Lysander-Crusader Equity Income F

Brandes Canadian Equity Class F

Thank You